UNIT 2

Data & Machine Learning Concepts

1. Concept Of Data And Information

What is Data?

→ Data refers to raw facts, figures, observations, measurements, symbols or records collected for a particular purpose.

→ Data by itself may not have a clear meaning until it is organized, processed and interpreted.

Examples of Data

- Student marks: 45, 67, 82, 91
- Names: Aman, Rahul, Simran
- Temperatures: 32°C, 35°C, 31°C
- Sales: ₹10,000, ₹15,000, ₹18,000
- Customer age: 21, 25, 32, 41
- Survey responses: Yes, No, Yes, Yes

Simple Definition:

Data = Raw facts and figures collected for processing.

Example

Suppose a college maintains:

Student	Age	Marks
A	19	78
B	20	85
C	19	72

These values are data because they are individual facts about students.

Characteristics of Data

Good data generally has the following characteristics:

1. **Accuracy** – Data should correctly represent the actual situation.
2. **Completeness** – Important values should not be missing.
3. **Consistency** – Data should follow the same format and rules.
4. **Relevance** – Data should be useful for the purpose for which it is collected.
5. **Timeliness** – Data should be available when required.
6. **Validity** – Data should follow predefined rules.
7. **Uniqueness** – Duplicate records should be avoided.

What is Information?

Information is processed, organized and meaningful data that helps a person understand a situation or make a decision.

In simple words:

Information = Processed Data + Meaning

Example

Raw data: 78, 85, 72, 90, 65

After processing:

- Average marks = 78
- Highest marks = 90
- Lowest marks = 65

→ The calculated results are information because they provide meaningful knowledge about student performance.

Data → Processing → Information → Decision

2. Types of Data (Part 1)

Based on the structure and organization, data is mainly classified into three types.



1 Structured Data

→ Structured data is organized in a fixed format (rows and columns).

→ It follows a predefined schema.

→ It is easy to store in relational databases and easy to analyze.

Examples

- ☑ Student database
- ☑ Bank transactions
- ☑ Employee records
- ☑ Sales records
- ☑ Inventory data
- ☑ Attendance records

Example

Student Table

Student_ID	Name	Age	Course	Marks
101	Aman	19	BBA	78
102	Simran	20	BBA	85
103	Rahul	19	BCA	72

Characteristics



- Fixed format
- Clearly defined schema
- Easy to search and analyze
- Stored in relational databases (RDBMS)
- Can be queried using SQL
- Highly organized

Quick Comparison at a Glance

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Structure	Fixed	Flexible	No fixed structure
Schema	Predefined	Partial / Self-describing	Usually absent
Format	Tables (rows & columns)	JSON, XML, HTML, Logs	Text, Images, Audio, Video
Storage	Relational Databases	NoSQL, XML/JSON DB	File systems, Data lakes, Object storage
Analysis	Easy	Moderate	More difficult
Example	Student marks table	JSON record, XML file	Videos, Emails, Images
Flexibility	Low	High	Very high

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2 Semi-Structured Data

→ Semi-structured data does not follow a rigid table format but contains tags, keys or metadata that provide some structure.

→ It lies between structured and unstructured data.



Examples

Example 1: JSON

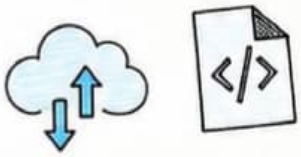
```
{
    "student_id": 101,
    "name": "Aman",
    "course": "BBA",
    "marks": 85
}
```

Example 2: XML

```
<student>
  <id>101</id>
  <name>Aman</name>
  <course>BBA</course>
  <marks>85</marks>
</student>
```

Characteristics

- Has some organizational structure (tags, keys, metadata).
- Does not require rigid table format.
- More flexible than structured data.
- Easier to analyze than unstructured data.
- Common in web applications, APIs, log files, NoSQL systems.



At a Glance

Structured	Semi-Structured	Unstructured
Fixed schema	Flexible schema	No fixed schema
Stored in tables	Uses tags/keys/metadata	Harder to organize
Easy to query	Moderate to analyze	Needs advanced tools
Examples: SQL DB, Excel	Examples: JSON, XML, Logs	Examples: Text, Images, Videos

3 Unstructured Data

→ Unstructured data does not have any predefined structure or format.

→ It is often generated in large quantities from various sources.

→ It may require special techniques like NLP, Computer Vision or Machine Learning for analysis.



"Unstructured data contains rich information but is difficult to organize.

Examples

- Emails
- Social media posts
- Images / Photos
- Videos
- Audio recordings
- PDF documents
- News articles
- Customer reviews
- Presentations

Characteristics

- No fixed structure or schema.
- Difficult to store in traditional databases.
- Requires preprocessing before analysis.
- Can be text, image, audio, video, etc.
- Needs advanced analytical tools and AI techniques.

Example

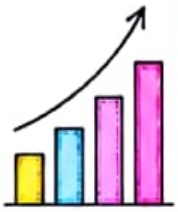
A customer writes:
"The product is excellent, but delivery was very slow and packaging was damaged."
This is unstructured text data.

3. Data Collection Methods

Data collection is the process of gathering relevant data for analysis and decision making.

Primary Data

- → Data collected first-hand by the researcher for a specific purpose.
- → Original and more reliable data.
- → Costly and time consuming.



Secondary Data

- → Data already collected by someone else for another purpose.
- → Easily available and less expensive.
- → May not perfectly fit current need.



Methods of Primary Data Collection

1. Questionnaire

- A set of questions given to respondents to collect data.
- Example: Online Google Form, survey sheet etc.

2. Interview

- Direct conversation with individuals to collect information.
- Example: Face to face interview, phone interview.

3. Observation

- Collecting data by watching and recording actual behaviour.
- Example: Observing customer behaviour in a store.

4. Experiment

- Data collected under controlled conditions by changing variables.
- Example: A/B testing in marketing.

5. Focus Group

- A small group discussion to gather opinions and ideas.
- Example: Discussion with selected customers or users.



Methods of Secondary Data

- ☑ Government reports & census data
- ☑ Research papers & journals
- ☑ Company annual reports
- ☑ Websites and online databases
- ☑ Books and articles
- ☑ Institutional records

Primary vs Secondary Data

Basis	Primary Data	Secondary Data
Source	Collected directly	Already collected
Purpose	For specific current study	For another purpose
Cost	Higher	Lower
Time	More time required	Less time required
Control	High control	Limited control
Example	Surveys, Interviews	Government reports, Websites

Key Takeaways

- ✓ Choose the right method based on the objective.
- ✓ Primary data is more specific and accurate.
- ✓ Secondary data is quick and economical.
- ✓ Often both primary and secondary data are used together.

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4. Data Quality and Preprocessing Basics

A. Data Quality

Data quality refers to how accurate, complete, consistent, reliable and useful data is for its intended purpose.

Good Quality Data
⇒ Better Analysis
⇒ Better Decisions

Poor quality data can lead to wrong analysis and poor decisions.

Dimensions of Data Quality

- **Accuracy** – Data should correctly represent the real world.
- **Completeness** – All required data should be available.
- **Consistency** – Data should be the same across all sources.
- **Validity** – Data should follow rules and constraints.
- **Uniqueness** – Duplicate records should be avoided.
- **Timeliness** – Data should be up-to-date and available when needed.
- **Relevance** – Data should be relevant to the objective.

Example

If a student's age is saved as 250 instead of 25, the data is not accurate and may lead to wrong calculations.

B. Data Preprocessing Basics

Data preprocessing is the process of cleaning, transforming and preparing raw data to make it suitable for analysis or machine learning models.

Steps in Data Preprocessing

1. Data Collection → 2. Data Cleaning → 3. Data Transformation → 4. Data Integration → 5. Data Reduction → 6. Ready for Analysis

Details of Important Preprocessing Steps

1. Data Cleaning
- Handling missing values
- Removing duplicates
- Correcting errors
- Handling outliers
- Standardizing formats

2. Handling Missing Values
- Remove the record
- Mean / Median imputation
- Mode imputation (for categorical)
- Forward / Backward fill
- Model based imputation

3. Handling Outliers
- Detect using methods like Z-score, IQR, Box plot
- Verify the outlier
- Correct or remove if it is an error
- Transform data if needed

4. Data Transformation
- Normalization / Standardization
- Encoding categorical data
- Log transformation
- Scaling numeric data
- Aggregation

5. Data Integration
- Combining data from multiple sources
- Matching using keys or IDs
- Resolving inconsistencies
- Creating a unified dataset

6. Data Reduction
- Sampling
- Feature selection
- Dimensionality reduction (PCA)
- Aggregation
- Removes unnecessary data

Well cleaned and preprocessed data leads to accurate analysis, reliable results and smart decisions.